

**CSA05 – Database Management Systems - Practical Examinations Model
Questions Paper**
Time: 3 Hours
MAX Marks: 100 (25 each)

SET -1

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Q1. Company Employee & Department Database (25 Marks)

Schema:

- DEPARTMENT (dept_id, dept_name, location)
- EMPLOYEE (emp_id, emp_name, dept_id, designation, salary, manager_id)
- PROJECT (proj_id, proj_name, dept_id)
- WORKS_ON (emp_id, proj_id, hours_worked)

Insert at least 6 departments, 12 employees (some with dept_id as NULL, i.e. unassigned), 5 projects, and 15 WORKS_ON records.

A. Joins (15 Marks)

1. Display employee name, department name, and designation using an **INNER JOIN** between EMPLOYEE and DEPARTMENT.
2. Using a **LEFT JOIN**, list all departments along with their employees, including departments that currently have no employees.
3. Using a **RIGHT JOIN**, list all employees along with their department name, including employees not yet assigned to any department.
4. Write a **SELF JOIN** on EMPLOYEE to display each employee's name alongside their manager's name.

B. Views (10 Marks)

1. Create a view EMP_DEPT_VIEW showing emp_name, dept_name, designation, salary.
2. Create a view HIGH_EARNERS showing employees earning above ₹50,000, using EMP_DEPT_VIEW as its base (a view built on a view).

Create a view DEPT_PROJECT_COUNT showing each department name along with the count of projects it runs.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.07 sec)
```

```
mysql> USE company_db;
Database changed
```

```
mysql> CREATE TABLE DEPARTMENT (
->   dept_id INT PRIMARY KEY,
->   dept_name VARCHAR(50),
->   location VARCHAR(50)
-> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql>
mysql> CREATE TABLE EMPLOYEE (
->   emp_id INT PRIMARY KEY,
->   emp_name VARCHAR(50),
->   dept_id INT NULL,
->   designation VARCHAR(50),
->   salary DECIMAL(10,2),
->   manager_id INT NULL,
->   FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> CREATE TABLE PROJECT (
->   proj_id INT PRIMARY KEY,
->   proj_name VARCHAR(50),
->   dept_id INT,
->   FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> CREATE TABLE WORKS_ON (
->   emp_id INT,
->   proj_id INT,
->   hours_worked INT,
->   PRIMARY KEY (emp_id, proj_id),
->   FOREIGN KEY (emp_id) REFERENCES EMPLOYEE(emp_id),
->   FOREIGN KEY (proj_id) REFERENCES PROJECT(proj_id)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_company_db |
+-----+
| department            |
| employee              |
| project               |
| works_on              |
+-----+
4 rows in set (0.02 sec)
```

```
mysql> INSERT INTO DEPARTMENT VALUES
-> (1, 'HR', 'Chennai'),
-> (2, 'IT', 'Bangalore'),
-> (3, 'Finance', 'Mumbai'),
-> (4, 'Marketing', 'Delhi'),
-> (5, 'Sales', 'Hyderabad'),
-> (6, 'Research', 'Pune');
Query OK, 6 rows affected (0.02 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM DEPARTMENT;
+-----+-----+-----+
| dept_id | dept_name | location |
+-----+-----+-----+
| 1       | HR       | Chennai  |
| 2       | IT       | Bangalore|
| 3       | Finance  | Mumbai   |
| 4       | Marketing| Delhi    |
| 5       | Sales    | Hyderabad|
| 6       | Research | Pune     |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```

mysql> INSERT INTO DEPARTMENT VALUES
-> (1, 'HR', 'Chennai'),
-> (2, 'IT', 'Bangalore'),
-> (3, 'Finance', 'Mumbai'),
-> (4, 'Marketing', 'Delhi'),
-> (5, 'Sales', 'Hyderabad'),
-> (6, 'Research', 'Pune');
ERROR 1062 (23000): Duplicate entry '1' for key 'department.PRIMARY'
mysql>
mysql> SELECT * FROM DEPARTMENT;
+-----+-----+-----+
| dept_id | dept_name | location |
+-----+-----+-----+
| 1 | HR | Chennai |
| 2 | IT | Bangalore |
| 3 | Finance | Mumbai |
| 4 | Marketing | Delhi |
| 5 | Sales | Hyderabad |
| 6 | Research | Pune |
+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> INSERT INTO EMPLOYEE VALUES
-> (101, 'Arun', 2, 'Software Engineer', 60000, NULL),
-> (102, 'Bala', 2, 'Developer', 55000, 101),
-> (103, 'Charan', 3, 'Accountant', 48000, NULL),
-> (104, 'Divya', 1, 'HR Executive', 45000, NULL),
-> (105, 'Esha', 4, 'Marketing Executive', 52000, NULL),
-> (106, 'Farhan', 5, 'Sales Executive', 47000, NULL),
-> (107, 'Gokul', 6, 'Research Analyst', 70000, NULL),
-> (108, 'Harish', NULL, 'Intern', 25000, 101),
-> (109, 'Isha', 2, 'Senior Developer', 85000, 101),
-> (110, 'Jaya', NULL, 'Consultant', 65000, NULL),
-> (111, 'Kiran', 5, 'Sales Manager', 75000, NULL),
-> (112, 'Lakshmi', 3, 'Finance Manager', 90000, NULL);
Query OK, 12 rows affected (0.01 sec)
Records: 12 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM EMPLOYEE;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | designation | salary | manager_id |
+-----+-----+-----+-----+-----+-----+
| 101 | Arun | 2 | Software Engineer | 60000.00 | NULL |
| 102 | Bala | 2 | Developer | 55000.00 | 101 |
| 103 | Charan | 3 | Accountant | 48000.00 | NULL |
| 104 | Divya | 1 | HR Executive | 45000.00 | NULL |
| 105 | Esha | 4 | Marketing Executive | 52000.00 | NULL |
| 106 | Farhan | 5 | Sales Executive | 47000.00 | NULL |
| 107 | Gokul | 6 | Research Analyst | 70000.00 | NULL |
| 108 | Harish | NULL | Intern | 25000.00 | 101 |
| 109 | Isha | 2 | Senior Developer | 85000.00 | 101 |
| 110 | Jaya | NULL | Consultant | 65000.00 | NULL |
| 111 | Kiran | 5 | Sales Manager | 75000.00 | NULL |
| 112 | Lakshmi | 3 | Finance Manager | 90000.00 | NULL |
+-----+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)

mysql> INSERT INTO PROJECT VALUES
-> (201, 'Payroll System', 3),
-> (202, 'Website Development', 2),
-> (203, 'Marketing Campaign', 4),
-> (204, 'Sales Automation', 5),
-> (205, 'AI Research', 6);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM PROJECT;
+-----+-----+-----+
| proj_id | proj_name | dept_id |
+-----+-----+-----+
| 201 | Payroll System | 3 |
| 202 | Website Development | 2 |
| 203 | Marketing Campaign | 4 |
| 204 | Sales Automation | 5 |
| 205 | AI Research | 6 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> INSERT INTO WORKS_ON VALUES
-> (101, 202, 40),
-> (102, 202, 35),
-> (103, 201, 30),
-> (104, 203, 25),
-> (105, 203, 40),
-> (106, 204, 35),
-> (107, 205, 45),
-> (109, 202, 40),
-> (111, 204, 30),
-> (112, 201, 35),
-> (101, 201, 20),
-> (102, 205, 15),
-> (105, 204, 20),
-> (107, 202, 10),

```

```
mysql>
mysql> SELECT * FROM WORKS_ON;
```

emp_id	proj_id	hours_worked
101	201	20
101	202	40
102	202	35
102	205	15
103	201	30
104	203	25
105	203	40
105	204	20
106	204	35
107	202	10
107	205	45
109	202	40
109	205	25
111	204	30
112	201	35

15 rows in set (0.00 sec)

```
mysql> SELECT
->     e.emp_name,
->     d.dept_name,
->     e.designation
-> FROM EMPLOYEE e
-> INNER JOIN DEPARTMENT d
-> ON e.dept_id = d.dept_id;
```

emp_name	dept_name	designation
Divya	HR	HR Executive
Arun	IT	Software Engineer
Bala	IT	Developer
Isha	IT	Senior Developer
Charan	Finance	Accountant
Lakshmi	Finance	Finance Manager
Esha	Marketing	Marketing Executive
Farhan	Sales	Sales Executive
Kiran	Sales	Sales Manager
Gokul	Research	Research Analyst

10 rows in set (0.00 sec)

```
mysql> SELECT
->     d.dept_name,
->     e.emp_name
-> FROM DEPARTMENT d
-> LEFT JOIN EMPLOYEE e
-> ON d.dept_id = e.dept_id;
```

dept_name	emp_name
HR	Divya
IT	Arun
IT	Bala
IT	Isha
Finance	Charan
Finance	Lakshmi
Marketing	Esha
Sales	Farhan
Sales	Kiran
Research	Gokul

10 rows in set (0.00 sec)

```
mysql> SELECT
->     e.emp_name,
->     d.dept_name
-> FROM DEPARTMENT d
-> RIGHT JOIN EMPLOYEE e
-> ON d.dept_id = e.dept_id;
```

emp_name	dept_name
Arun	IT
Bala	IT
Charan	Finance
Divya	HR
Esha	Marketing
Farhan	Sales
Gokul	Research
Harish	NULL
Isha	IT
Jaya	NULL
Kiran	Sales
Lakshmi	Finance

12 rows in set (0.00 sec)

```
mysql> SELECT
->     e.emp_name AS Employee,
->     m.emp_name AS Manager
-> FROM EMPLOYEE e
-> LEFT JOIN EMPLOYEE m
-> ON e.manager_id = m.emp_id;
```

Employee	Manager
Arun	NULL
Bala	Arun
Charan	NULL
Divya	NULL
Esha	NULL
Farhan	NULL
Gokul	NULL
Harish	Arun
Isha	Arun
Jaya	NULL
Kiran	NULL
Lakshmi	NULL

12 rows in set (0.00 sec)

```
mysql> CREATE VIEW EMP_DEPT_VIEW AS
-> SELECT
->     e.emp_name,
->     d.dept_name,
->     e.designation,
->     e.salary
-> FROM EMPLOYEE e
-> INNER JOIN DEPARTMENT d
-> ON e.dept_id = d.dept_id;
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> SELECT * FROM EMP_DEPT_VIEW;
```

emp_name	dept_name	designation	salary
Divya	HR	HR Executive	45000.00
Arun	IT	Software Engineer	60000.00
Bala	IT	Developer	55000.00
Isha	IT	Senior Developer	85000.00
Charan	Finance	Accountant	40000.00
Lakshmi	Finance	Finance Manager	90000.00
Esha	Marketing	Marketing Executive	52000.00
Farhan	Sales	Sales Executive	47000.00
Kiran	Sales	Sales Manager	75000.00
Gokul	Research	Research Analyst	70000.00

10 rows in set (0.00 sec)

```
mysql> CREATE VIEW HIGH_EARNERS AS
-> SELECT *
-> FROM EMP_DEPT_VIEW
-> WHERE salary > 50000;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM HIGH_EARNERS;
```

emp_name	dept_name	designation	salary
Arun	IT	Software Engineer	60000.00
Bala	IT	Developer	55000.00
Esha	Marketing	Marketing Executive	52000.00
Gokul	Research	Research Analyst	70000.00
Isha	IT	Senior Developer	85000.00
Kiran	Sales	Sales Manager	75000.00
Lakshmi	Finance	Finance Manager	90000.00

7 rows in set (0.00 sec)

```
mysql>
mysql> CREATE VIEW DEPT_PROJECT_COUNT AS
-> SELECT
->     d.dept_name,
->     COUNT(p.proj_id) AS project_count
-> FROM DEPARTMENT d
-> LEFT JOIN PROJECT p
-> ON d.dept_id = p.dept_id
-> GROUP BY d.dept_id, d.dept_name;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM DEPT_PROJECT_COUNT;
```

dept_name	project_count
HR	0
IT	1
Finance	1
Marketing	1
Sales	1
Research	1

6 rows in set (0.01 sec)

FINAL OUTPUT :

```
mysql> SHOW TABLES;
+-----+
| Tables_in_company_db |
+-----+
| department            |
| dept_project_count    |
| emp_dept_view         |
| employee              |
| high_earners          |
| project               |
| works_on              |
+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM EMP_DEPT_VIEW;
+-----+-----+-----+-----+
| emp_name | dept_name | designation | salary |
+-----+-----+-----+-----+
| Divya    | HR        | HR Executive | 45000.00 |
| Arun     | IT        | Software Engineer | 60000.00 |
| Bala     | IT        | Developer    | 55000.00 |
| Isha     | IT        | Senior Developer | 85000.00 |
| Charan   | Finance   | Accountant   | 48000.00 |
| Lakshmi  | Finance   | Finance Manager | 90000.00 |
| Esha     | Marketing | Marketing Executive | 52000.00 |
| Farhan   | Sales     | Sales Executive | 47000.00 |
| Kiran    | Sales     | Sales Manager | 75000.00 |
| Gokul    | Research  | Research Analyst | 70000.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM HIGH_EARNERS;
+-----+-----+-----+-----+
| emp_name | dept_name | designation | salary |
+-----+-----+-----+-----+
| Arun     | IT        | Software Engineer | 60000.00 |
| Bala     | IT        | Developer    | 55000.00 |
| Esha     | Marketing | Marketing Executive | 52000.00 |
| Gokul    | Research  | Research Analyst | 70000.00 |
| Isha     | IT        | Senior Developer | 85000.00 |
| Kiran    | Sales     | Sales Manager | 75000.00 |
| Lakshmi  | Finance   | Finance Manager | 90000.00 |
+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM DEPT_PROJECT_COUNT;
+-----+-----+
| dept_name | project_count |
+-----+-----+
| HR        | 0 |
| IT        | 1 |
| Finance   | 1 |
| Marketing | 1 |
| Sales     | 1 |
| Research  | 1 |
+-----+-----+
6 rows in set (0.00 sec)
```

Q2. University Student, Course & Faculty Database (25 Marks)

Schema:

- STUDENT (student_id, student_name, dept)
- FACULTY (faculty_id, faculty_name, dept)
- COURSE (course_id, course_name, faculty_id, credits)
- ENROLLMENT (enroll_id, student_id, course_id, marks)

Insert at least 8 students, 5 faculty, 6 courses, and 15 enrollment records (include at least one course with no enrollments, and one student not enrolled in any course).

A. Joins (15 Marks)

1. Display student name, course name, and marks using an **INNER JOIN** across STUDENT, ENROLLMENT, and COURSE.
2. Using a **LEFT JOIN**, list all students along with the courses they are enrolled in, including students who are enrolled in no course.
3. Using a **RIGHT JOIN** (or equivalent), list all courses along with enrolled student names, including courses with zero enrollments.
4. Find faculty members who teach courses in which no student has scored above 80 marks (join + aggregate + HAVING).

B. Views (10 Marks)

1. Create a view STUDENT_COURSE_VIEW displaying student_name, course_name, faculty_name, marks.
2. Create a view TOPPERS showing students who scored more than 90 marks in any course, derived from STUDENT_COURSE_VIEW.
3. Create a view FACULTY_WORKLOAD showing faculty name and the number of courses they teach, using GROUP BY inside the view.
4. Query FACULTY_WORKLOAD to find the faculty member with the maximum workload.



```
mysql>
mysql> CREATE DATABASE university_db;
Query OK, 1 row affected (0.02 sec)

mysql> USE university_db;
Database changed
mysql> CREATE TABLE STUDENT (
  ->     student_id INT PRIMARY KEY,
  ->     student_name VARCHAR(50),
  ->     dept VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE FACULTY (
  ->     faculty_id INT PRIMARY KEY,
  ->     faculty_name VARCHAR(50),
  ->     dept VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE COURSE (
  ->     course_id INT PRIMARY KEY,
  ->     course_name VARCHAR(50),
  ->     faculty_id INT,
  ->     credits INT,
  ->     FOREIGN KEY (faculty_id) REFERENCES FACULTY(faculty_id)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE ENROLLMENT (
  ->     enroll_id INT PRIMARY KEY,
  ->     student_id INT,
  ->     course_id INT,
  ->     marks INT,
  ->     FOREIGN KEY (student_id) REFERENCES STUDENT(student_id),
  ->     FOREIGN KEY (course_id) REFERENCES COURSE(course_id)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO STUDENT VALUES
  -> (1, 'Aarav', 'CSE'),
  -> (2, 'Bhavya', 'ECE'),
  -> (3, 'Charan', 'CSE'),
  -> (4, 'Divya', 'EEE'),
  -> (5, 'Eshan', 'CSE'),
  -> (6, 'Farah', 'ECE'),
  -> (7, 'Gowtham', 'MECH'),
  -> (8, 'Harini', 'CSE');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> INSERT INTO FACULTY VALUES
  -> (101, 'Dr. Kumar', 'CSE'),
  -> (102, 'Dr. Priya', 'ECE'),
  -> (103, 'Dr. Ravi', 'EEE'),
  -> (104, 'Dr. Meena', 'MECH'),
  -> (105, 'Dr. Anil', 'CSE');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> INSERT INTO COURSE VALUES
  -> (201, 'Database Management', 101, 4),
  -> (202, 'Data Structures', 105, 4),
  -> (203, 'Computer Networks', 101, 3),
  -> (204, 'Digital Electronics', 102, 3),
  -> (205, 'Engineering Mathematics', 103, 4),
  -> (206, 'Machine Design', 104, 3);
Query OK, 6 rows affected (0.02 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> INSERT INTO ENROLLMENT VALUES
  -> (1, 1, 201, 95),
  -> (2, 1, 202, 88),
  -> (3, 2, 204, 76),
  -> (4, 2, 201, 82),
  -> (5, 3, 201, 91),
  -> (6, 3, 203, 85),
  -> (7, 4, 205, 79),
  -> (8, 4, 204, 72),
  -> (9, 5, 202, 94),
  -> (10, 5, 203, 89),
  -> (11, 6, 204, 81),
  -> (12, 6, 205, 87),
  -> (13, 8, 201, 96),
  -> (14, 8, 202, 93),
  -> (15, 8, 203, 90);
Query OK, 15 rows affected (0.04 sec)
Records: 15 Duplicates: 0 Warnings: 0
```



```
mysql> SELECT * FROM STUDENT;
```

student_id	student_name	dept
1	Aarav	CSE
2	Bhavya	ECE
3	Charan	CSE
4	Divya	EEE
5	Eshan	CSE
6	Farah	ECE
7	Gowtham	MECH
8	Harini	CSE

```
8 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM FACULTY;
```

faculty_id	faculty_name	dept
101	Dr. Kumar	CSE
102	Dr. Priya	ECE
103	Dr. Ravi	EEE
104	Dr. Meena	MECH
105	Dr. Anil	CSE

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM COURSE;
```

course_id	course_name	faculty_id	credits
201	Database Management	101	4
202	Data Structures	105	4
203	Computer Networks	101	3
204	Digital Electronics	102	3
205	Engineering Mathematics	103	4
206	Machine Design	104	3

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ENROLLMENT;
```

enroll_id	student_id	course_id	marks
1	1	201	95
2	1	202	88
3	2	204	76
4	2	201	82
5	3	201	91
6	3	203	85
7	4	205	79
8	4	204	72
9	5	202	94
10	5	203	89
11	6	204	81
12	6	205	87
13	8	201	96
14	8	202	93
15	8	203	90

```
15 rows in set (0.00 sec)
```

```
mysql> SELECT
-> s.student_name,
-> c.course_name,
-> e.marks
-> FROM STUDENT s
-> INNER JOIN ENROLLMENT e
-> ON s.student_id = e.student_id
-> INNER JOIN COURSE c
-> ON e.course_id = c.course_id;
```

student_name	course_name	marks
Aarav	Database Management	95
Bhavya	Database Management	82
Charan	Database Management	91
Harini	Database Management	96
Aarav	Data Structures	88
Eshan	Data Structures	94
Harini	Data Structures	93
Charan	Computer Networks	85
Eshan	Computer Networks	89
Harini	Computer Networks	90
Bhavya	Digital Electronics	76
Divya	Digital Electronics	72
Farah	Digital Electronics	81
Divya	Engineering Mathematics	79
Farah	Engineering Mathematics	87

```
15 rows in set (0.00 sec)
```

```
mysql> SELECT
->     s.student_name,
->     c.course_name,
->     e.marks
-> FROM STUDENT s
-> LEFT JOIN ENROLLMENT e
-> ON s.student_id = e.student_id
-> LEFT JOIN COURSE c
-> ON e.course_id = c.course_id;
```

student_name	course_name	marks
Aarav	Database Management	95
Aarav	Data Structures	88
Bhavya	Digital Electronics	76
Bhavya	Database Management	82
Charan	Database Management	91
Charan	Computer Networks	85
Divya	Engineering Mathematics	79
Divya	Digital Electronics	72
Eshan	Data Structures	94
Eshan	Computer Networks	89
Farah	Digital Electronics	81
Farah	Engineering Mathematics	87
Gowtham	NULL	NULL
Harini	Database Management	96
Harini	Data Structures	93
Harini	Computer Networks	90

16 rows in set (0.00 sec)

```
mysql> SELECT
->     c.course_name,
->     s.student_name,
->     e.marks
-> FROM STUDENT s
-> RIGHT JOIN ENROLLMENT e
-> ON s.student_id = e.student_id
-> RIGHT JOIN COURSE c
-> ON e.course_id = c.course_id;
```

course_name	student_name	marks
Database Management	Aarav	95
Database Management	Bhavya	82
Database Management	Charan	91
Database Management	Harini	96
Data Structures	Aarav	88
Data Structures	Eshan	94
Data Structures	Harini	93
Computer Networks	Charan	85
Computer Networks	Eshan	89
Computer Networks	Harini	90
Digital Electronics	Bhavya	76
Digital Electronics	Divya	72
Digital Electronics	Farah	81
Engineering Mathematics	Divya	79
Engineering Mathematics	Farah	87
Machine Design	NULL	NULL

16 rows in set (0.00 sec)

```
mysql> SELECT
->     f.faculty_name
-> FROM FACULTY f
-> INNER JOIN COURSE c
-> ON f.faculty_id = c.faculty_id
-> INNER JOIN ENROLLMENT e
-> ON c.course_id = e.course_id
-> GROUP BY f.faculty_id, f.faculty_name
-> HAVING MAX(e.marks) <= 80;
```

Empty set (0.00 sec)

```
mysql> CREATE VIEW STUDENT_COURSE_VIEW AS
-> SELECT
->     s.student_name,
->     c.course_name,
->     f.faculty_name,
->     e.marks
-> FROM STUDENT s
-> INNER JOIN ENROLLMENT e
-> ON s.student_id = e.student_id
-> INNER JOIN COURSE c
-> ON e.course_id = c.course_id
-> INNER JOIN FACULTY f
-> ON c.faculty_id = f.faculty_id;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM STUDENT_COURSE_VIEW;
```

student_name	course_name	faculty_name	marks
Aarav	Database Management	Dr. Kumar	95
Bhavya	Database Management	Dr. Kumar	82
Charan	Database Management	Dr. Kumar	91
Harini	Database Management	Dr. Kumar	96
Charan	Computer Networks	Dr. Kumar	85
Eshan	Computer Networks	Dr. Kumar	89
Harini	Computer Networks	Dr. Kumar	90
Bhavya	Digital Electronics	Dr. Priya	76
Divya	Digital Electronics	Dr. Priya	72
Farah	Digital Electronics	Dr. Priya	81
Divya	Engineering Mathematics	Dr. Ravi	79
Farah	Engineering Mathematics	Dr. Ravi	87
Aarav	Data Structures	Dr. Anil	88
Eshan	Data Structures	Dr. Anil	94
Harini	Data Structures	Dr. Anil	93

```
15 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE VIEW TOPPERS AS
-> SELECT *
-> FROM STUDENT_COURSE_VIEW
-> WHERE marks > 90;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM TOPPERS;
```

student_name	course_name	faculty_name	marks
Aarav	Database Management	Dr. Kumar	95
Charan	Database Management	Dr. Kumar	91
Harini	Database Management	Dr. Kumar	96
Eshan	Data Structures	Dr. Anil	94
Harini	Data Structures	Dr. Anil	93

```
5 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW FACULTY_WORKLOAD AS
-> SELECT
->     f.faculty_name,
->     COUNT(c.course_id) AS course_count
-> FROM FACULTY f
-> LEFT JOIN COURSE c
-> ON f.faculty_id = c.faculty_id
-> GROUP BY f.faculty_id, f.faculty_name;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE VIEW FACULTY_WORKLOAD AS
-> SELECT
->     f.faculty_name,
->     COUNT(c.course_id) AS course_count
-> FROM FACULTY f
-> LEFT JOIN COURSE c
-> ON f.faculty_id = c.faculty_id
->
->
-> ^C
```

```
mysql> SELECT * FROM FACULTY_WORKLOAD;
```

faculty_name	course_count
Dr. Kumar	2
Dr. Priya	1
Dr. Ravi	1
Dr. Meena	1
Dr. Anil	1

```
5 rows in set (0.01 sec)
```

```
mysql> SELECT *
-> FROM FACULTY_WORKLOAD
-> WHERE course_count = (
->     SELECT MAX(course_count)
->     FROM FACULTY_WORKLOAD
-> );
```

faculty_name	course_count
Dr. Kumar	2

```
1 row in set (0.00 sec)
```

FINAL OUTPUT :

```
mysql> SHOW TABLES;
+-----+
| Tables_in_university_db |
+-----+
| course                    |
| enrollment                |
| faculty                   |
| faculty_workload           |
| student                   |
| student_course_view        |
| toppers                    |
+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM STUDENT_COURSE_VIEW;
+-----+-----+-----+-----+
| student_name | course_name | faculty_name | marks |
+-----+-----+-----+-----+
| Aarav        | Database Management | Dr. Kumar    | 95    |
| Bhavya       | Database Management | Dr. Kumar    | 82    |
| Charan       | Database Management | Dr. Kumar    | 91    |
| Harini       | Database Management | Dr. Kumar    | 96    |
| Charan       | Computer Networks   | Dr. Kumar    | 85    |
| Eshan        | Computer Networks   | Dr. Kumar    | 89    |
| Harini       | Computer Networks   | Dr. Kumar    | 90    |
| Bhavya       | Digital Electronics  | Dr. Priya    | 76    |
| Divya        | Digital Electronics  | Dr. Priya    | 72    |
| Farah        | Digital Electronics  | Dr. Priya    | 81    |
| Divya        | Engineering Mathematics | Dr. Ravi     | 79    |
| Farah        | Engineering Mathematics | Dr. Ravi     | 87    |
| Aarav        | Data Structures     | Dr. Anil     | 88    |
| Eshan        | Data Structures     | Dr. Anil     | 94    |
| Harini       | Data Structures     | Dr. Anil     | 93    |
+-----+-----+-----+-----+
15 rows in set (0.00 sec)

mysql> SELECT * FROM TOPPERS;
+-----+-----+-----+-----+
| student_name | course_name | faculty_name | marks |
+-----+-----+-----+-----+
| Aarav        | Database Management | Dr. Kumar    | 95    |
| Charan       | Database Management | Dr. Kumar    | 91    |
| Harini       | Database Management | Dr. Kumar    | 96    |
| Eshan        | Data Structures     | Dr. Anil     | 94    |
| Harini       | Data Structures     | Dr. Anil     | 93    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT * FROM FACULTY_WORKLOAD;
+-----+-----+
| faculty_name | course_count |
+-----+-----+
| Dr. Kumar    | 2            |
| Dr. Priya    | 1            |
| Dr. Ravi     | 1            |
| Dr. Meena    | 1            |
| Dr. Anil     | 1            |
+-----+-----+
5 rows in set (0.00 sec)
```

Q3. Hospital Patient, Doctor & Appointment Database (25 Marks)

Schema:

- DOCTOR (doctor_id, doctor_name, specialization, dept_id)
- PATIENT (patient_id, patient_name, age, gender)
- APPOINTMENT (appt_id, patient_id, doctor_id, appt_date, status)
- DEPARTMENT (dept_id, dept_name)

Insert at least 5 departments, 8 doctors, 10 patients, and 15 appointments (include a patient with no appointments and a doctor with no appointments).

A. Joins (15 Marks)

1. Display patient name, doctor name, and appointment date using an **INNER JOIN** between PATIENT, APPOINTMENT, and DOCTOR.
2. Using a **LEFT JOIN**, list all patients with their appointments (if any), including patients who have never booked an appointment.
3. Using a **RIGHT JOIN**, list all doctors with their appointments, including doctors with no appointments scheduled.
4. Display doctor name, specialization, and department name using a join between DOCTOR and DEPARTMENT.

B. Views (10 Marks)

1. Create a view APPOINTMENT_DETAILS showing patient_name, doctor_name, specialization, appt_date, status.
2. Create a view DOCTOR_APPOINTMENT_COUNT showing doctor name and total number of appointments handled, using GROUP BY.
3. Create a view PENDING_APPOINTMENTS showing only appointments where status = 'Pending', based on APPOINTMENT_DETAILS.
4. Use PENDING_APPOINTMENTS to update (via the base table) the status of a specific appointment to 'Completed', and verify the view reflects the change.



```
mysql>
mysql>
mysql> CREATE DATABASE hospital_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE hospital_db;CREATE TABLE DEPARTMENT (
Database changed
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> CREATE TABLE DOCTOR (
-> doctor_id INT PRIMARY KEY,
-> doctor_name VARCHAR(50),
-> specialization VARCHAR(50),
-> dept_id INT,
-> FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE PATIENT (
-> patient_id INT PRIMARY KEY,
-> patient_name VARCHAR(50),
-> age INT,
-> gender VARCHAR(10)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE APPOINTMENT (
-> appt_id INT PRIMARY KEY,
-> patient_id INT,
-> doctor_id INT,
-> appt_date DATE,
-> status VARCHAR(20),
-> FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id),
-> FOREIGN KEY (doctor_id) REFERENCES DOCTOR(doctor_id)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO DEPARTMENT VALUES
-> (1, 'Cardiology'),
-> (2, 'Neurology'),
-> (3, 'Orthopedics'),
-> (4, 'Pediatrics'),
-> (5, 'Dermatology');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> INSERT INTO DOCTOR VALUES
-> (101, 'Dr. Sharma', 'Cardiologist', 1),
-> (102, 'Dr. Priya', 'Neurologist', 2),
-> (103, 'Dr. Ramesh', 'Orthopedic', 3),
-> (104, 'Dr. Anitha', 'Pediatrician', 4),
-> (105, 'Dr. Kumar', 'Dermatologist', 5),
-> (106, 'Dr. Meena', 'Cardiologist', 1),
-> (107, 'Dr. Raj', 'Neurologist', 2),
-> (108, 'Dr. Sneha', 'Orthopedic', 3);
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> INSERT INTO PATIENT VALUES
-> (1, 'Arjun', 35, 'Male'),
-> (2, 'Bhavya', 28, 'Female'),
-> (3, 'Charan', 45, 'Male'),
-> (4, 'Divya', 12, 'Female'),
-> (5, 'Eshan', 52, 'Male'),
-> (6, 'Farah', 31, 'Female'),
-> (7, 'Gokul', 60, 'Male'),
-> (8, 'Harini', 24, 'Female'),
-> (9, 'Ishaan', 40, 'Male'),
-> (10, 'Jaya', 67, 'Female');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO APPOINTMENT VALUES
-> (1001, 1, 101, '2026-09-01', 'Completed'),
-> (1002, 2, 102, '2026-09-01', 'Pending'),
-> (1003, 3, 103, '2026-09-02', 'Completed'),
-> (1004, 4, 104, '2026-09-02', 'Pending'),
-> (1005, 5, 105, '2026-09-03', 'Completed'),
-> (1006, 6, 106, '2026-09-03', 'Pending'),
-> (1007, 7, 107, '2026-09-04', 'Completed'),
-> (1008, 8, 101, '2026-09-04', 'Pending'),
-> (1009, 9, 102, '2026-09-05', 'Completed'),
-> (1010, 1, 103, '2026-09-05', 'Pending'),
-> (1011, 2, 104, '2026-09-06', 'Completed'),
-> (1012, 3, 105, '2026-09-06', 'Pending'),
-> (1013, 5, 106, '2026-09-07', 'Completed'),
-> (1014, 6, 101, '2026-09-07', 'Pending'),
-> (1015, 8, 107, '2026-09-08', 'Completed');
Query OK, 15 rows affected (0.01 sec)
Records: 15 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT
-> p.patient_name,
-> d.doctor_name,
-> a.appt_date
-> FROM PATIENT p
-> INNER JOIN APPOINTMENT a
-> ON p.patient_id = a.patient_id
-> INNER JOIN DOCTOR d
-> ON a.doctor_id = d.doctor_id;
```

patient_name	doctor_name	appt_date
Arjun	Dr. Sharma	2026-09-01
Harini	Dr. Sharma	2026-09-04
Farah	Dr. Sharma	2026-09-07
Bhavya	Dr. Priya	2026-09-01
Ishaan	Dr. Priya	2026-09-05
Charan	Dr. Ramesh	2026-09-02
Arjun	Dr. Ramesh	2026-09-05
Divya	Dr. Anitha	2026-09-02
Bhavya	Dr. Anitha	2026-09-06
Eshan	Dr. Kumar	2026-09-03
Charan	Dr. Kumar	2026-09-06
Farah	Dr. Meena	2026-09-03
Eshan	Dr. Meena	2026-09-07
Gokul	Dr. Raj	2026-09-04
Harini	Dr. Raj	2026-09-08

15 rows in set (0.00 sec)

```
mysql> SELECT
-> p.patient_name,
-> a.appt_date,
-> a.status
-> FROM PATIENT p
-> LEFT JOIN APPOINTMENT a
-> ON p.patient_id = a.patient_id;
```

patient_name	appt_date	status
Arjun	2026-09-01	Completed
Arjun	2026-09-05	Pending
Bhavya	2026-09-01	Pending
Bhavya	2026-09-06	Completed
Charan	2026-09-02	Completed
Charan	2026-09-06	Pending
Divya	2026-09-02	Pending
Eshan	2026-09-03	Completed
Eshan	2026-09-07	Completed
Farah	2026-09-03	Pending
Farah	2026-09-07	Pending
Gokul	2026-09-04	Completed
Harini	2026-09-04	Pending
Harini	2026-09-08	Completed
Ishaan	2026-09-05	Completed
Jaya	NULL	NULL

16 rows in set (0.00 sec)

```
mysql> SELECT
-> d.doctor_name,
-> a.appt_date,
-> a.status
-> FROM APPOINTMENT a
-> RIGHT JOIN DOCTOR d
-> ON a.doctor_id = d.doctor_id;
```

doctor_name	appt_date	status
Dr. Sharma	2026-09-01	Completed
Dr. Sharma	2026-09-04	Pending
Dr. Sharma	2026-09-07	Pending
Dr. Priya	2026-09-01	Pending
Dr. Priya	2026-09-05	Completed
Dr. Ramesh	2026-09-02	Completed
Dr. Ramesh	2026-09-05	Pending
Dr. Anitha	2026-09-02	Pending
Dr. Anitha	2026-09-06	Completed
Dr. Kumar	2026-09-03	Completed
Dr. Kumar	2026-09-06	Pending
Dr. Meena	2026-09-03	Pending
Dr. Meena	2026-09-07	Completed
Dr. Raj	2026-09-04	Completed
Dr. Raj	2026-09-08	Completed
Dr. Sneha	NULL	NULL

16 rows in set (0.00 sec)

```
mysql> SELECT
-> d.doctor_name,
-> d.specialization,
-> dept.dept_name
-> FROM DOCTOR d
-> INNER JOIN DEPARTMENT dept
-> ON d.dept_id = dept.dept_id;
```

```
mysql> SELECT
->     d.doctor_name,
->     d.specialization,
->     dept.dept_name
-> FROM DOCTOR d
-> INNER JOIN DEPARTMENT dept
-> ON d.dept_id = dept.dept_id;
```

doctor_name	specialization	dept_name
Dr. Sharma	Cardiologist	Cardiology
Dr. Meena	Cardiologist	Cardiology
Dr. Priya	Neurologist	Neurology
Dr. Raj	Neurologist	Neurology
Dr. Ramesh	Orthopedic	Orthopedics
Dr. Sneha	Orthopedic	Orthopedics
Dr. Anitha	Pediatrician	Pediatrics
Dr. Kumar	Dermatologist	Dermatology

```
8 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW APPOINTMENT_DETAILS AS
-> SELECT
->     p.patient_name,
->     d.doctor_name,
->     d.specialization,
->     a.appt_date,
->     a.status
-> FROM APPOINTMENT a
-> INNER JOIN PATIENT p
-> ON a.patient_id = p.patient_id
-> INNER JOIN DOCTOR d
-> ON a.doctor_id = d.doctor_id;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM APPOINTMENT_DETAILS;
```

patient_name	doctor_name	specialization	appt_date	status
Arjun	Dr. Sharma	Cardiologist	2026-09-01	Completed
Harini	Dr. Sharma	Cardiologist	2026-09-04	Pending
Farah	Dr. Sharma	Cardiologist	2026-09-07	Pending
Bhavya	Dr. Priya	Neurologist	2026-09-01	Pending
Ishaan	Dr. Priya	Neurologist	2026-09-05	Completed
Charan	Dr. Ramesh	Orthopedic	2026-09-02	Completed
Arjun	Dr. Ramesh	Orthopedic	2026-09-05	Pending
Divya	Dr. Anitha	Pediatrician	2026-09-02	Pending
Bhavya	Dr. Anitha	Pediatrician	2026-09-06	Completed
Eshan	Dr. Kumar	Dermatologist	2026-09-03	Completed
Charan	Dr. Kumar	Dermatologist	2026-09-06	Pending
Farah	Dr. Meena	Cardiologist	2026-09-03	Pending
Eshan	Dr. Meena	Cardiologist	2026-09-07	Completed
Gokul	Dr. Raj	Neurologist	2026-09-04	Completed
Harini	Dr. Raj	Neurologist	2026-09-08	Completed

```
15 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW DOCTOR_APPOINTMENT_COUNT AS
-> SELECT
->     d.doctor_name,
->     COUNT(a.appt_id) AS total_appointments
-> FROM DOCTOR d
-> LEFT JOIN APPOINTMENT a
-> ON d.doctor_id = a.doctor_id
-> GROUP BY d.doctor_id, d.doctor_name;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM DOCTOR_APPOINTMENT_COUNT;
```

doctor_name	total_appointments
Dr. Sharma	3
Dr. Priya	2
Dr. Ramesh	2
Dr. Anitha	2
Dr. Kumar	2
Dr. Meena	2
Dr. Raj	2
Dr. Sneha	0

```
8 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW PENDING_APPOINTMENTS AS
-> SELECT *
-> FROM APPOINTMENT_DETAILS
-> WHERE status = 'Pending';
```

Query OK, 0 rows affected (0.01 sec)

FINAL OUTPUT :

```
mysql>
mysql> SHOW TABLES;
+-----+
| Tables_in_hospital_db |
+-----+
| appointment            |
| appointment_details     |
| department             |
| doctor                 |
| doctor_appointment_count |
| patient                |
| pending_appointments    |
+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM APPOINTMENT_DETAILS;
+-----+-----+-----+-----+-----+
| patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+
| Arjun        | Dr. Sharma  | Cardiologist   | 2026-09-01 | Completed |
| Harini       | Dr. Sharma  | Cardiologist   | 2026-09-04 | Pending  |
| Farah        | Dr. Sharma  | Cardiologist   | 2026-09-07 | Pending  |
| Bhavya       | Dr. Priya   | Neurologist    | 2026-09-01 | Completed |
| Ishaan       | Dr. Priya   | Neurologist    | 2026-09-05 | Completed |
| Charan       | Dr. Ramesh  | Orthopedic     | 2026-09-02 | Completed |
| Arjun        | Dr. Ramesh  | Orthopedic     | 2026-09-05 | Pending  |
| Divya        | Dr. Anitha  | Pediatrician   | 2026-09-02 | Pending  |
| Bhavya       | Dr. Anitha  | Pediatrician   | 2026-09-06 | Completed |
| Eshan        | Dr. Kumar   | Dermatologist  | 2026-09-03 | Completed |
| Charan       | Dr. Kumar   | Dermatologist  | 2026-09-06 | Pending  |
| Farah        | Dr. Meena   | Cardiologist   | 2026-09-03 | Pending  |
| Eshan        | Dr. Meena   | Cardiologist   | 2026-09-07 | Completed |
| Gokul        | Dr. Raj     | Neurologist    | 2026-09-04 | Completed |
| Harini       | Dr. Raj     | Neurologist    | 2026-09-08 | Completed |
+-----+-----+-----+-----+-----+
15 rows in set (0.00 sec)

mysql> SELECT * FROM DOCTOR_APPOINTMENT_COUNT;
+-----+-----+
| doctor_name | total_appointments |
+-----+-----+
| Dr. Sharma  | 3 |
| Dr. Priya   | 2 |
| Dr. Ramesh  | 2 |
| Dr. Anitha  | 2 |
| Dr. Kumar   | 2 |
| Dr. Meena   | 2 |
| Dr. Raj     | 2 |
| Dr. Sneha   | 0 |
+-----+-----+
8 rows in set (0.00 sec)

mysql> SELECT * FROM PENDING_APPOINTMENTS;
+-----+-----+-----+-----+-----+
| patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+
| Divya        | Dr. Anitha  | Pediatrician   | 2026-09-02 | Pending |
| Farah        | Dr. Meena   | Cardiologist   | 2026-09-03 | Pending |
| Harini       | Dr. Sharma  | Cardiologist   | 2026-09-04 | Pending |
| Arjun        | Dr. Ramesh  | Orthopedic     | 2026-09-05 | Pending |
| Charan       | Dr. Kumar   | Dermatologist  | 2026-09-06 | Pending |
| Farah        | Dr. Sharma  | Cardiologist   | 2026-09-07 | Pending |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Q4. E-Commerce Customer, Order & Product Database (25 Marks)

Schema:

- CUSTOMER (cust_id, cust_name, city)
- PRODUCT (prod_id, prod_name, category, price)
- ORDERS (order_id, cust_id, order_date)
- ORDER_ITEMS (order_id, prod_id, quantity)

Insert at least 6 customers, 10 products, 10 orders, and 20 order-item records (include one customer with no orders and one product never ordered).

A. Joins (15 Marks)

1. Display customer name, order date, product name, and quantity using an **INNER JOIN** across all four tables.
2. Using a **LEFT JOIN**, list all customers with their orders, including customers who have never placed an order.
3. Using a **RIGHT JOIN** (or equivalent), list all products with the orders they appear in, including products never ordered.
4. Find the top-selling product category by total revenue, using multi-table joins with GROUP BY and ORDER BY.

B. Views (10 Marks)

1. Create a view ORDER_SUMMARY showing order_id, cust_name, order_date, total_amount (computed using joins and SUM).
2. Create a view CUSTOMER_SPENDING showing customer name and total amount spent across all orders, using GROUP BY inside the view.
3. Query CUSTOMER_SPENDING to find customers who have spent more than ₹10,000 in total.
4. Create a view PRODUCT_SALES showing product name, category, and total quantity sold, then use it to find the least-sold product.



```
mysql>
mysql> CREATE DATABASE ecommerce_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE ecommerce_db;
Database changed
mysql> CREATE TABLE CUSTOMER (
  ->     cust_id INT PRIMARY KEY,
  ->     cust_name VARCHAR(50),
  ->     city VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE PRODUCT (
  ->     prod_id INT PRIMARY KEY,
  ->     prod_name VARCHAR(50),
  ->     category VARCHAR(50),
  ->     price DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE ORDERS (
  ->     order_id INT PRIMARY KEY,
  ->     cust_id INT,
  ->     order_date DATE,
  ->     FOREIGN KEY (cust_id) REFERENCES CUSTOMER(cust_id)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE ORDER_ITEMS (
  ->     order_id INT,
  ->     prod_id INT,
  ->     quantity INT,
  ->     PRIMARY KEY (order_id, prod_id),
  ->     FOREIGN KEY (order_id) REFERENCES ORDERS(order_id),
  ->     FOREIGN KEY (prod_id) REFERENCES PRODUCT(prod_id)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO CUSTOMER VALUES
  -> (1, 'Arjun', 'Chennai'),
  -> (2, 'Bhavya', 'Bangalore'),
  -> (3, 'Charan', 'Hyderabad'),
  -> (4, 'Divya', 'Mumbai'),
  -> (5, 'Eshan', 'Delhi'),
  -> (6, 'Harini', 'Pune');
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> INSERT INTO PRODUCT VALUES
  -> (101, 'Laptop', 'Electronics', 55000),
  -> (102, 'Smartphone', 'Electronics', 30000),
  -> (103, 'Headphones', 'Electronics', 5000),
  -> (104, 'Keyboard', 'Accessories', 2500),
  -> (105, 'Mouse', 'Accessories', 1500),
  -> (106, 'Office Chair', 'Furniture', 12000),
  -> (107, 'Study Table', 'Furniture', 15000),
  -> (108, 'Backpack', 'Fashion', 3000),
  -> (109, 'Shoes', 'Fashion', 4000),
  -> (110, 'Smart Watch', 'Electronics', 8000);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO ORDERS VALUES
  -> (1001, 1, '2026-08-01'),
  -> (1002, 2, '2026-08-02'),
  -> (1003, 3, '2026-08-03'),
  -> (1004, 4, '2026-08-04'),
  -> (1005, 1, '2026-08-05'),
  -> (1006, 2, '2026-08-06'),
  -> (1007, 3, '2026-08-07'),
  -> (1008, 4, '2026-08-08'),
  -> (1009, 5, '2026-08-09'),
  -> (1010, 6, '2026-08-10');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO ORDER_ITEMS VALUES
-> (1001, 101, 1),
-> (1001, 103, 2),
->
-> (1002, 102, 1),
-> (1002, 105, 2),
->
-> (1003, 106, 1),
-> (1003, 108, 2),
->
-> (1004, 107, 1),
-> (1004, 109, 2),
->
-> (1005, 101, 1),
-> (1005, 104, 2),
->
-> (1006, 102, 2),
-> (1006, 103, 1),
->
-> (1007, 106, 1),
-> (1007, 105, 3),
->
-> (1008, 107, 1),
-> (1008, 108, 1),
->
-> (1009, 101, 1),
-> (1009, 109, 2),
->
-> (1010, 102, 1),
-> (1010, 106, 1);
Query OK, 20 rows affected (0.01 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT
-> c.cust_name,
-> o.order_date,
-> p.prod_name,
-> oi.quantity
-> FROM CUSTOMER c
-> INNER JOIN ORDERS o
-> ON c.cust_id = o.cust_id
-> INNER JOIN ORDER_ITEMS oi
-> ON o.order_id = oi.order_id
-> INNER JOIN PRODUCT p
-> ON oi.prod_id = p.prod_id;
+-----+-----+-----+-----+
| cust_name | order_date | prod_name | quantity |
+-----+-----+-----+-----+
| Arjun | 2026-08-01 | Laptop | 1 |
| Arjun | 2026-08-01 | Headphones | 2 |
| Arjun | 2026-08-05 | Laptop | 1 |
| Arjun | 2026-08-05 | Keyboard | 2 |
| Bhavya | 2026-08-02 | Smartphone | 1 |
| Bhavya | 2026-08-02 | Mouse | 2 |
| Bhavya | 2026-08-06 | Smartphone | 2 |
| Bhavya | 2026-08-06 | Headphones | 1 |
| Charan | 2026-08-03 | Office Chair | 1 |
| Charan | 2026-08-03 | Backpack | 2 |
| Charan | 2026-08-07 | Mouse | 3 |
| Charan | 2026-08-07 | Office Chair | 1 |
| Divya | 2026-08-04 | Study Table | 1 |
| Divya | 2026-08-04 | Shoes | 2 |
| Divya | 2026-08-08 | Study Table | 1 |
| Divya | 2026-08-08 | Backpack | 1 |
| Eshan | 2026-08-09 | Laptop | 1 |
| Eshan | 2026-08-09 | Shoes | 2 |
| Eshan | 2026-08-10 | Smartphone | 1 |
| Eshan | 2026-08-10 | Office Chair | 1 |
+-----+-----+-----+-----+
20 rows in set (0.00 sec)
```

```
mysql> SELECT
-> c.cust_name,
-> o.order_id,
-> o.order_date
-> FROM CUSTOMER c
-> LEFT JOIN ORDERS o
-> ON c.cust_id = o.cust_id;
+-----+-----+-----+
| cust_name | order_id | order_date |
+-----+-----+-----+
| Arjun | 1001 | 2026-08-01 |
| Arjun | 1005 | 2026-08-05 |
| Bhavya | 1002 | 2026-08-02 |
| Bhavya | 1006 | 2026-08-06 |
| Charan | 1003 | 2026-08-03 |
| Charan | 1007 | 2026-08-07 |
| Divya | 1004 | 2026-08-04 |
| Divya | 1008 | 2026-08-08 |
| Eshan | 1009 | 2026-08-09 |
| Eshan | 1010 | 2026-08-10 |
| Harini | NULL | NULL |
+-----+-----+-----+
11 rows in set (0.00 sec)
```

```
mysql> SELECT
->   p.prod_name,
->   o.order_id,
->   oi.quantity
-> FROM ORDERS o
-> RIGHT JOIN ORDER_ITEMS oi
-> ON o.order_id = oi.order_id
-> RIGHT JOIN PRODUCT p
-> ON oi.prod_id = p.prod_id;
+-----+-----+-----+
| prod_name | order_id | quantity |
+-----+-----+-----+
| Laptop    | 1001     | 1         |
| Laptop    | 1005     | 1         |
| Laptop    | 1009     | 1         |
| Smartphone | 1002     | 1         |
| Smartphone | 1006     | 2         |
| Smartphone | 1010     | 1         |
| Headphones | 1001     | 2         |
| Headphones | 1006     | 1         |
| Keyboard   | 1005     | 2         |
| Mouse      | 1002     | 2         |
| Mouse      | 1007     | 3         |
| Office Chair | 1003     | 1         |
| Office Chair | 1007     | 1         |
| Office Chair | 1010     | 1         |
| Study Table | 1004     | 1         |
| Study Table | 1008     | 1         |
| Backpack   | 1003     | 2         |
| Backpack   | 1008     | 1         |
| Shoes      | 1004     | 2         |
| Shoes      | 1009     | 2         |
| Smart Watch | NULL     | NULL      |
+-----+-----+-----+
21 rows in set (0.00 sec)

mysql> SELECT
->   p.category,
->   SUM(p.price * oi.quantity) AS total_revenue
-> FROM PRODUCT p
-> INNER JOIN ORDER_ITEMS oi
-> ON p.prod_id = oi.prod_id
-> GROUP BY p.category
-> ORDER BY total_revenue DESC
-> LIMIT 1;
+-----+-----+
| category | total_revenue |
+-----+-----+
| Electronics | 300000.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE VIEW ORDER_SUMMARY AS
-> SELECT
->   o.order_id,
->   c.cust_name,
->   o.order_date,
->   SUM(p.price * oi.quantity) AS total_amount
-> FROM ORDERS o
-> INNER JOIN CUSTOMER c
-> ON o.cust_id = c.cust_id
-> INNER JOIN ORDER_ITEMS oi
-> ON o.order_id = oi.order_id
-> INNER JOIN PRODUCT p
-> ON oi.prod_id = p.prod_id
-> GROUP BY o.order_id, c.cust_name, o.order_date;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM ORDER_SUMMARY;
+-----+-----+-----+-----+
| order_id | cust_name | order_date | total_amount |
+-----+-----+-----+-----+
| 1001     | Arjun    | 2026-08-01 | 65000.00     |
| 1005     | Arjun    | 2026-08-05 | 60000.00     |
| 1002     | Bhavya   | 2026-08-02 | 33000.00     |
| 1006     | Bhavya   | 2026-08-06 | 65000.00     |
| 1003     | Charan   | 2026-08-03 | 18000.00     |
| 1007     | Charan   | 2026-08-07 | 16500.00     |
| 1004     | Divya    | 2026-08-04 | 23000.00     |
| 1008     | Divya    | 2026-08-08 | 18000.00     |
| 1009     | Eshan    | 2026-08-09 | 63000.00     |
| 1010     | Eshan    | 2026-08-10 | 42000.00     |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> CREATE VIEW CUSTOMER_SPENDING AS
-> SELECT
->   c.cust_name,
->   COALESCE(SUM(p.price * oi.quantity), 0) AS total_spent
-> FROM CUSTOMER c
-> LEFT JOIN ORDERS o
-> ON c.cust_id = o.cust_id
-> LEFT JOIN ORDER_ITEMS oi
-> ON o.order_id = oi.order_id
```

```
mysql> SELECT * FROM CUSTOMER_SPENDING;
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Arjun     | 125000.00 |
| Bhavya    | 98000.00  |
| Charan    | 34500.00  |
| Divya     | 41000.00  |
| Eshan     | 105000.00 |
| Harini    | 0.00      |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT *
-> FROM CUSTOMER_SPENDING
-> WHERE total_spent > 10000;
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Arjun     | 125000.00 |
| Bhavya    | 98000.00  |
| Charan    | 34500.00  |
| Divya     | 41000.00  |
| Eshan     | 105000.00 |
+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE VIEW PRODUCT_SALES AS
-> SELECT
->   p.prod_name,
->   p.category,
->   COALESCE(SUM(oi.quantity), 0) AS total_quantity_
-> FROM PRODUCT p
-> LEFT JOIN ORDER_ITEMS oi
-> ON p.prod_id = oi.prod_id
-> GROUP BY p.prod_id, p.prod_name, p.category;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM PRODUCT_SALES;
+-----+-----+-----+
| prod_name | category | total_quantity_sold |
+-----+-----+-----+
| Laptop    | Electronics | 3 |
| Smartphone | Electronics | 4 |
| Headphones | Electronics | 3 |
| Keyboard  | Accessories | 2 |
| Mouse     | Accessories | 5 |
| Office Chair | Furniture | 3 |
| Study Table | Furniture | 2 |
| Backpack  | Fashion | 3 |
| Shoes     | Fashion | 4 |
| Smart Watch | Electronics | 0 |
+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> SELECT *
-> FROM PRODUCT_SALES
-> WHERE total_quantity_sold = (
->   SELECT MIN(total_quantity_sold)
->   FROM PRODUCT_SALES
-> );
+-----+-----+-----+
| prod_name | category | total_quantity_sold |
+-----+-----+-----+
| Smart Watch | Electronics | 0 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

FINAL OUTPUT :

```
mysql>
mysql>
mysql> SHOW TABLES;
+-----+
| Tables_in_ecommerce_db |
+-----+
| customer                 |
| customer_spending        |
| order_items              |
| order_summary            |
| orders                   |
| product                  |
| product_sales            |
+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM ORDER_SUMMARY;
+-----+-----+-----+-----+
| order_id | cust_name | order_date | total_amount |
+-----+-----+-----+-----+
| 1001     | Arjun    | 2026-08-01 | 65000.00     |
| 1005     | Arjun    | 2026-08-05 | 60000.00     |
| 1002     | Bhavya   | 2026-08-02 | 33000.00     |
| 1006     | Bhavya   | 2026-08-06 | 65000.00     |
| 1003     | Charan   | 2026-08-03 | 18000.00     |
| 1007     | Charan   | 2026-08-07 | 16500.00     |
| 1004     | Divya    | 2026-08-04 | 23000.00     |
| 1008     | Divya    | 2026-08-08 | 18000.00     |
| 1009     | Eshan    | 2026-08-09 | 63000.00     |
| 1010     | Eshan    | 2026-08-10 | 42000.00     |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM CUSTOMER_SPENDING;
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Arjun     | 125000.00  |
| Bhavya    | 98000.00   |
| Charan    | 34500.00   |
| Divya     | 41000.00   |
| Eshan     | 105000.00  |
| Harini    | 0.00       |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM PRODUCT_SALES;
+-----+-----+-----+
| prod_name | category | total_quantity_sold |
+-----+-----+-----+
| Laptop    | Electronics | 3 |
| Smartphone | Electronics | 4 |
| Headphones | Electronics | 3 |
| Keyboard  | Accessories | 2 |
| Mouse     | Accessories | 5 |
| Office Chair | Furniture | 3 |
| Study Table | Furniture | 2 |
| Backpack  | Fashion | 3 |
| Shoes     | Fashion | 4 |
| Smart Watch | Electronics | 0 |
+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT *
-> FROM CUSTOMER_SPENDING
-> WHERE total_spent > 10000;
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Arjun     | 125000.00  |
| Bhavya    | 98000.00   |
| Charan    | 34500.00   |
| Divya     | 41000.00   |
| Eshan     | 105000.00  |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT *
-> FROM PRODUCT_SALES
-> WHERE total_quantity_sold = (
->     SELECT MIN(total_quantity_sold)
->     FROM PRODUCT_SALES
-> );
+-----+-----+-----+
| prod_name | category | total_quantity_sold |
+-----+-----+-----+
| Smart Watch | Electronics | 0 |
+-----+-----+-----+
1 row in set (0.00 sec)
```